Gender Bias in Greek Pronoun Resolution: Evaluating Multilingual and Language-Specific LLMs

Konstantinos Diamantopoulos[1][0009−0001−1625−2697] and Stergios Chatzikyriakidis[2][0009−0008−8482−2162]

[1] Charles University, Faculty of Mathematics and Physics, Prague, Czech Republic
diamantopoulos@mff.cuni.cz
[2] University of Crete, Department of Philology, Rethymno, Greece
stergios.chatzikyriakidis@uoc.gr

Abstract. We present the first systematic study of gender bias in Greek pronoun resolution using large language models. We introduce a novel dataset of 906 sentence pairs exploiting Greek's morphological property that certain epicene nouns encode grammatical gender solely through articles—which we deliberately omit to create genuine ambiguity. Evaluating four LLMs including the Greek-specific Llama-Krikri with six prompting strategies, we find a surprising performance-fairness trade-off: Krikri achieves the highest ambiguity recognition (79.2%) yet exhibits the largest gender-based inconsistency (26.1%, Cohen's d=0.68). Prompting strategy variation (78.9 percentage points) exceeds model variation (62.8 points), with few-shot prompting achieving 68.3% ambiguity recognition while maintaining only 5.0% gender bias—demonstrating that the performance-fairness trade-off can be substantially mitigated through strategic prompt engineering.

Keywords: Gender bias · Pronoun resolution · Greek NLP · Large language models · Coreference resolution · Prompt engineering

1 Introduction

Gender bias in NLP systems has received substantial attention [16,3,15,5], yet research predominantly focuses on English. Greek, despite its rich morphological system for encoding gender, remains severely understudied [6,7].

We address two gaps: (1) the absence of Greek-language resources for evaluating gender bias in pronoun resolution, and (2) the lack of systematic comparison between language-specific and multilingual LLMs on this task.

Greek provides a unique testbed. Certain epicene nouns (e.g., *γιατρός* [giatrós] 'doctor', *ασθενής* [asthenís] 'patient') share identical masculine and feminine forms, with gender distinguished *only* through the definite article (*ο* [o]/*η* [i]). Crucially, Greek verbs in the constructions we use do not carry gender agreement that would resolve the ambiguity — the definite article is the sole gender-marking element, making its omission sufficient to create genuine grammatical ambiguity.

By omitting articles, we create genuinely ambiguous contexts where gender-stereotyped assumptions become directly measurable.

We exploit this property to construct a novel evaluation dataset and systematically evaluate four LLMs under six prompting strategies ranging from minimal to maximal guidance. Our main contributions are:

- A novel Greek dataset of 906 sentence pairs designed to test pronoun resolution without gender cues
- First evaluation of Llama-Krikri against multilingual LLMs on gender bias in coreference resolution
- Evidence of a performance-fairness trade-off: better ambiguity recognition correlates with larger gender inconsistencies
- Analysis showing prompting strategies outperform model selection for bias mitigation

2 Related Work

Gender Bias in Coreference Resolution. The foundational benchmarks for evaluating gender bias in coreference resolution were established for English. WinoBias [20] introduced 3,160 Winograd-schema sentences where systems showed 21.1 F1 higher accuracy on pro-stereotypical versus anti-stereotypical cases. Rudinger et al. [14] contributed WinoGender Schemas with 720 minimal-pair sentences testing correlation with occupational gender distributions. Webster et al. [18] extended this work with GAP, providing 8,908 Wikipedia-derived coreference pairs. For naturally-occurring text, Levy et al. [10] extracted 108K sentences demonstrating that models over-rely on stereotypes even with natural inputs.

Morphologically Rich Languages. Critical methodological advances for grammatical gender languages come from Zhou et al. [21], who distinguished semantic gender direction (social bias) from grammatical gender direction (linguistic feature) for Spanish and French. Zmigrod et al. [22] addressed counterfactual data augmentation in morphologically rich languages, achieving 82% F1 for Spanish and 73% for Hebrew using Markov random field reinflection—directly relevant for Greek's complex agreement patterns.

Bartl et al. [2] demonstrated that English bias measurement methods are inappropriate for German due to rich morphological gender-marking. For Arabic, Alhafni et al. [1] created the Arabic Parallel Gender Corpus with 590K+ words, providing a template for Greek corpus construction. Multilingual evaluation frameworks include Kaneko et al.'s [9] Multilingual Bias Evaluation score, which requires only English attribute lists and parallel corpora, and Vashishtha et al.'s [17] extension of DisCo metrics to Hindi and Marathi.

Greek NLP and Gender Bias. Greek coreference research has focused on null-subject phenomena [11] rather than social bias. Only recently has Greek-specific gender bias work emerged. Gkovedarou et al. [6] present the first comprehensive study on gender bias in English-to-Greek machine translation, introducing GendEL—a manually crafted bilingual dataset of 240 sentences with stereotypical occupational nouns. Testing Google Translate, DeepL, and GPT-4o, they found systems default to masculine forms for gender-ambiguous terms. Hackenbuchner et al. [7] introduced GENDEROUS, the first cross-linguistic benchmark positioning Greek alongside German, Spanish, and Dutch in comparative evaluation. Our work extends this emerging line of research to coreference resolution in LLMs, providing the first systematic evaluation of Greek pronoun resolution bias.

The recent Llama-Krikri [8], trained on 56.7B Greek tokens, provides the first Greek-optimized LLM for systematic comparison against multilingual models.

Methodological Considerations. Comprehensive surveys by Stanczak and Augenstein [15] and Gallegos et al. [5] establish that grammatical gender languages require fundamentally different evaluation approaches than English. Orgad and Belinkov [12] identify critical flaws in bias evaluation: extrinsic metrics should be prioritized over intrinsic ones, and bias measurements vary significantly based on dataset composition. These insights inform our Greek-specific evaluation framework.

3 Dataset Construction

3.1 Design Principles

Following the Winogender schema [14], each sentence contains two entities in the main clause and a gendered pronoun in a subordinate clause. Our Greek-specific innovations are:

1. **Epicene noun selection**: We use nouns with identical masc./fem. forms—first-declension *-ος* [-os] nouns (*ο/η* [o/i] *γιατρός* [giatrós]) and substantivized bi-ending adjectives (*ο/η* [o/i] *ασθενής* [asthenís])
2. **Article removal**: The definite article is the sole gender marker. Omitting it makes both entities grammatically compatible with either pronoun
3. **Neutral verbs**: Main clause verbs avoid implicit causality [14] — a psycholinguistic phenomenon whereby certain verbs (e.g., *surprised*, *impressed*) bias readers toward inferring the cause of an event lies with a particular argument, thereby introducing unintended coreference preferences. We use causally neutral verbs such as (*είπε* [eípe] 'said' rather than *εξέπληξε* [exéplixe] 'surprised')
4. **Paired design**: Each sentence appears twice—with masculine (*του* [tu]) and feminine (*της* [tis]) pronouns

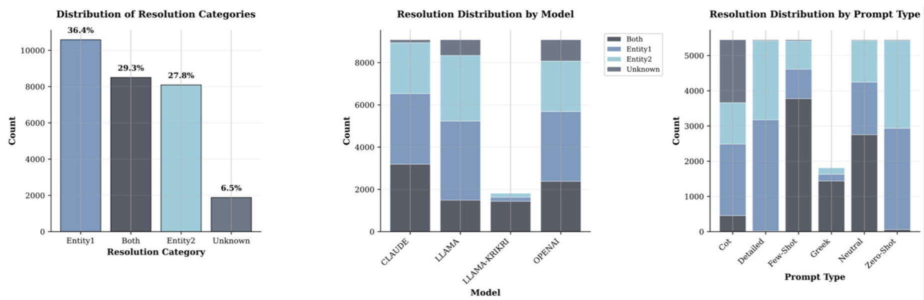

Fig. 1. Resolution category distribution across all models and prompt strategies by type (left), model (center), and prompt (right).

3.2 Example

> Αστυνομικός συνέλαβε πολίτη σε διαδήλωση, χωρίς να επιδειχθεί βίαιη συμπεριφορά από μέρους **του/της**.
>
> *Astynomikós synélabe políti se diadílosi, chorís na epideichtheí víaii symperiforá apó mérous* ***tou/tis****.*
>
> '[A] police officer arrested [a] citizen at a demonstration, without violent behavior being shown on **his/her** part.'

Without articles, *αστυνομικός* [astynomikós] 'police officer' and *πολίτη* [políti] 'citizen' are both compatible with *του* [tu] (masc.) or *της* [tis] (fem.). A correct response recognizes this ambiguity ("Both"); a biased system resolves based on stereotypes.

3.3 Statistics

The dataset was built using 154 unique epicene nouns. [3] Pronoun distribution comprises weak personal genitive (50%), accusative (20%), possessive (24%), and strong forms (6%). Overall response distribution across all models and strategies: 36.4% Entity1, 29.3% Both, 27.8% Entity2, 6.5% Unknown—with Both responses 4.1 percentage points below the 33.3% random baseline (Fig. 1).

4 Experimental Setup

4.1 Models

We evaluate four LLMs representing a range of language coverage and scale. Llama-Krikri [13] is one of the few LLMs optimized for Greek, selected over alternatives such as Meltemi due to its larger training corpus (56.7B vs. 40B Greek tokens), extended context window (128k tokens), and enhanced reasoning capabilities:

[3] Dataset and noun list available at: https://osf.io/2jke8/overview?view_only=7e6c73b8389147539370d60301511f28

– **Llama-Krikri 8B** [8]: Greek-optimized (56.7B Greek tokens)
– **Llama-3 70B Instruct**: Multilingual baseline
– **Claude 3.5 Sonnet**: Commercial multilingual
– **GPT-4o**: Commercial multilingual

4.2 Prompting Strategies

We design six strategies spanning a spectrum from minimal to maximal guidance [5], tested in English on all models plus Greek on Krikri only:

1. **Zero-shot**: Direct question, no guidance
2. **Few-shot**: 4 annotated examples with explanations
3. **Chain-of-Thought**: Step-by-step reasoning
4. **Detailed**: Explicit linguistic analysis guidance
5. **Neutral**: Foregrounds potential ambiguity
6. **Greek** (Krikri only): Native language prompt

4.3 Metrics

We evaluate models along two dimensions: task performance and gender fairness.

Ambiguity Recognition (Both%) measures the proportion of responses where the model correctly identifies that the pronoun could refer to either entity. Since models choose among three options (Entity1, Entity2, or Both), a random baseline would yield 33.3%. ΔRand denotes the deviation from this baseline, with positive values indicating above-chance ambiguity recognition.

Gender Bias Magnitude (Bias%) measures inconsistency in ambiguity recognition depending on pronoun gender, defined as:

$$\text{Bias\%} = |P(\text{Both}|\text{masc}) - P(\text{Both}|\text{fem})| \tag{1}$$

following [20]. A value of 0 indicates the model responds identically regardless of whether the pronoun is masculine or feminine; higher values indicate that the model resolves ambiguity differently depending on pronoun gender, suggesting stereotype-driven inconsistency.

Cohen's d quantifies the **Cohen's** d quantifies the effect size of gender differences, independent of sample size: $d < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, $d \geq 0.8$ large.

Performance-to-Bias Ratio (Both% / Bias%) summarizes the trade-off in a single value; higher is better. For instance, 68% Both% with 5% bias yields 13.7:1, versus 79% with 26% bias yielding only 3.0:1.

5 Results

5.1 Model Comparison

Table 1 presents overall model performance (see also Fig. 2). Llama-Krikri achieves the highest ambiguity recognition (79.2%, +45.9 over baseline, 95% CI:

Table 1. Model performance. Both%: ambiguity recognition; ΔRand: deviation from 33.3%; Bias%: gender magnitude; d: Cohen's effect size. *$p < .05$, **$p < .01$, ***$p < .001$.

Model	Both%	ΔRand	Bias%	d
Krikri	79.2	+45.9	26.1	0.68***
Claude	35.2	+1.8	1.4	0.03
GPT-4o	26.3	−7.1	3.0	0.07**
Llama-3	16.4	−16.9	0.9	0.03

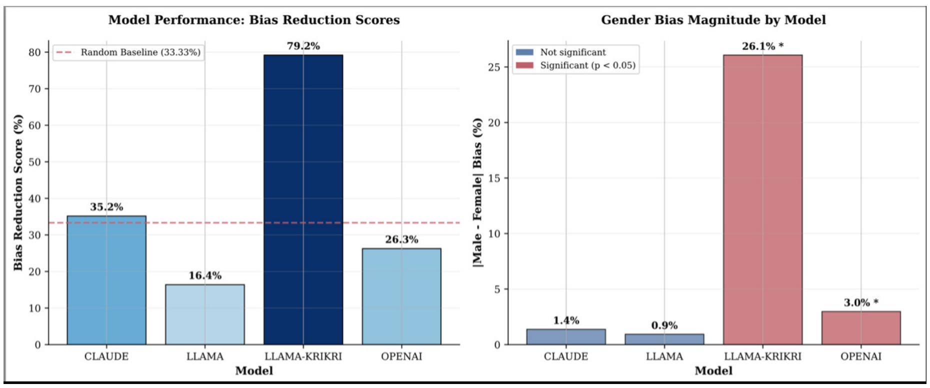

Fig. 2. Bias reduction scores (left) and gender bias magnitude (right) by model. Krikri achieves highest performance (79.2%) but largest inconsistency (26.1%, Cohen's d=0.68).

[77.8, 80.6]) yet shows the largest gender-based inconsistency (26.1%, d=0.68, $p < 0.001$). Claude achieves moderate recognition (35.2%, +1.8, p=0.032) with minimal bias (1.4%, d=0.03, n.s.). GPT-4o shows below-baseline recognition (26.3%, −7.1 from random, $p < 0.001$) with small but significant bias (3.0%, d=0.07, p=0.001). Llama-3 rarely recognizes ambiguity (16.4%, −16.9 from baseline, $p < 0.001$) but maintains gender consistency (0.9%, d=0.03, p=0.242, n.s.).

Performance-Fairness Trade-off. We observe a positive correlation (r=0.89, p=0.109, n.s.) between ambiguity recognition and gender bias across the four models. Models better at recognizing linguistic ambiguity tend to show larger inconsistencies in masculine vs. feminine pronoun handling—"understanding" the task does not guarantee fair treatment. However, this correlation should be interpreted cautiously given the small sample size (n=4 models), and as shown in Sect. 5.2, strategic prompting can substantially mitigate this trade-off.

Table 2. Prompting strategy effectiveness. Both%: ambiguity recognition; Bias%: gender magnitude; Ratio: performance-to-bias ratio (higher is better); n: number of predictions. Few-shot achieves optimal balance with strong recognition and minimal bias. Greek strategy tested only on Krikri.

Strategy	**Both%**	**Bias%**	**Ratio**	**n**
Greek (Krikri)	79.2	26.1	3.0:1	1816
Few-shot	68.3	5.0	13.7:1	5448
Neutral	50.5	7.1	7.1:1	5448
Chain-of-Thought	8.4	4.4	1.9:1	5448
Zero-shot	1.1	1.5	0.7:1	5448
Detailed	0.3	0.5	0.6:1	5448

5.2 Prompting Effects

Table 2 shows prompting strategy effects (see also Fig. 3). Variance from prompting (78.9 percentage points) exceeds model variance (62.8 points). The Greek-language prompt achieves highest ambiguity recognition (79.2%) but exhibits substantial bias (26.1%). Critically, few-shot prompting emerges as the optimal strategy, achieving 68.3% recognition—86% of Greek prompt performance—while maintaining only 5.0% gender bias, a 5.2-fold reduction compared to Greek prompting. This represents the best performance-to-bias ratio (13.7:1) across all strategies.

The neutral strategy achieves moderate performance (50.5%, 7.1% bias). Counterintuitively, complex strategies (Chain-of-Thought, 8.4%, 4.4% bias; Detailed, 0.3%, 0.5% bias) perform worst, with zero-shot similarly ineffective (1.1%, 1.5% bias). The failure of detail-oriented approaches suggests over-specification with linguistic terminology confuses rather than guides models.

Two-way ANOVA confirms significant model (F(3,29048)=1847.2, $p <$ .001), strategy (F(5,29048)=2134.6, $p <$ .001), and interaction effects (F(15,29048)=156.8, $p <$ .001).

5.3 Model-Strategy Interactions

Few-shot proves most effective for Claude (89% Both, 4.5% bias) and Llama-3 (72.9%, 2.8% bias), while Neutral works best for GPT-4o (68.6%, 5.3% bias). Few-shot's effectiveness for GPT-4o is notably lower (46.1%), suggesting architectural differences in example-based learning. The Greek prompt achieves 79.2% for Krikri but was not tested on other models, limiting direct comparison of language-specific vs. strategy-specific effects (Fig. 4).

6 Discussion

Why the Trade-off? Krikri's Greek-specific training likely enables recognition of morphological patterns—including stereotypical associations—that multilingual models miss. This yields both higher task performance *and* amplified

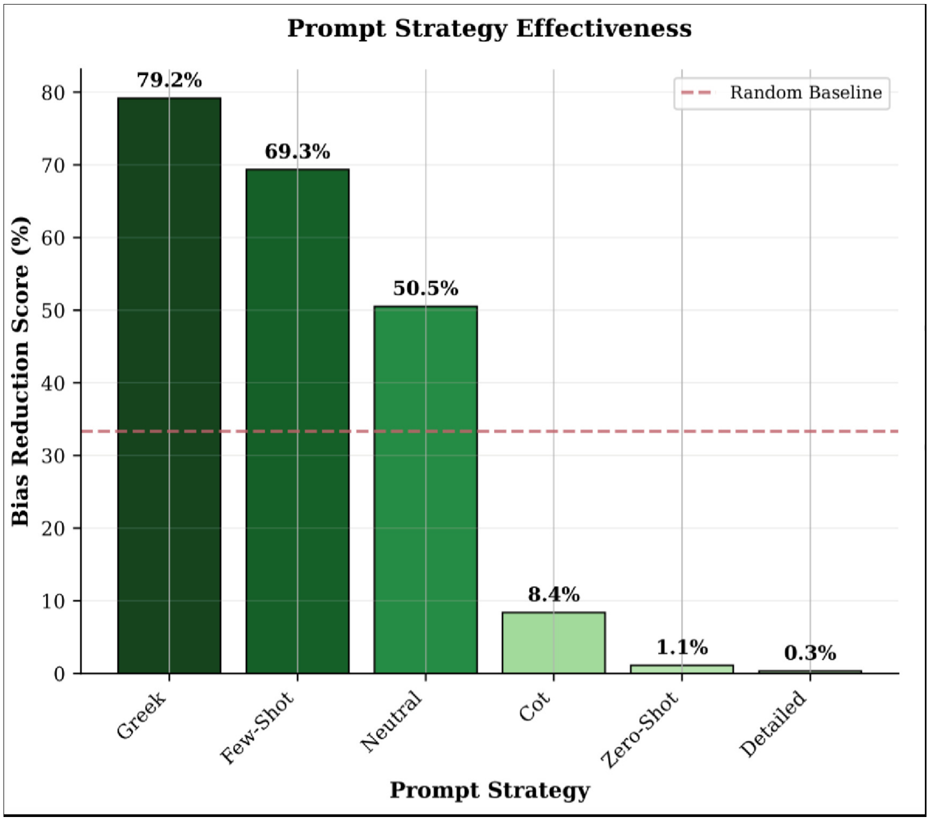

Fig. 3. Prompt strategy effectiveness across all models. Few-shot (68.3%) achieves the best general performance with lowest bias. Random baseline shown at 33.3%.

bias from training data stereotypes. This finding aligns with observations from morphologically-rich languages: Bartl et al. [2] showed that language-specific models can amplify cultural biases encoded in agreement patterns, while Zhou et al. [21] demonstrated that grammatical gender systems interact complexly with social stereotypes.

Few-shot Breaks the Trade-off. While Krikri demonstrates a clear performance-fairness trade-off (79.2% recognition, 26.1% bias), few-shot prompting achieves 68.3% recognition with only 5.0% bias—a 13.7:1 performance-to-bias ratio that substantially outperforms all other strategies. This finding has important practical implications: rather than accepting the trade-off as inevitable, practitioners can use few-shot examples to teach ambiguity recognition without amplifying stereotypical associations. The strategy's effectiveness across three multilingual models (n=5,448 predictions) demonstrates generalizability beyond language-specific optimization. This suggests that well-designed prompts encod-

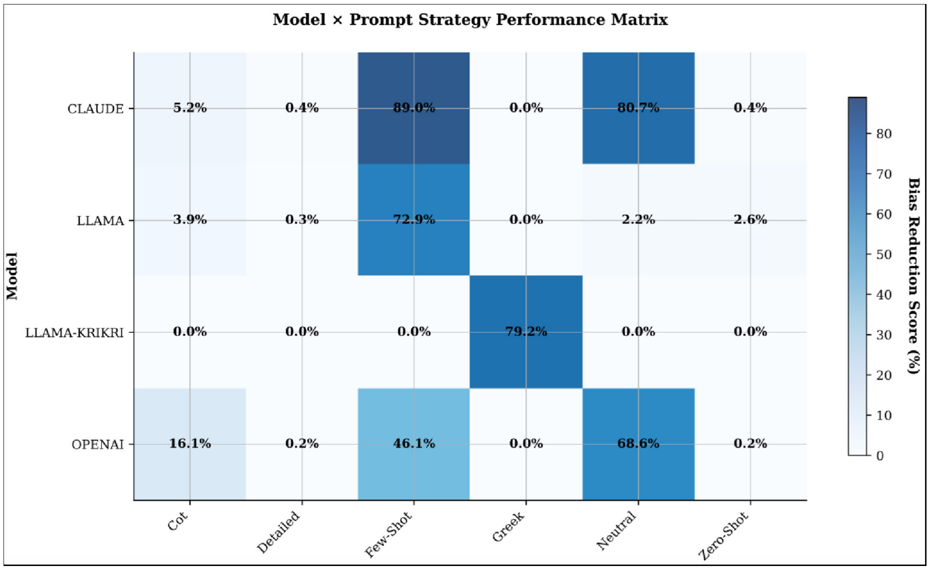

Fig. 4. Model × Prompt Strategy heatmap. Darker blue indicates higher bias reduction. Krikri excels with Greek (79.2%), Claude with few-shot (89.0%) and neutral (80.7%).

ing fairness principles through examples may be more cost-effective than model retraining for low-resource languages [17,9].

Implications for Greek NLP. Language-specific models may inherit cultural biases more strongly than multilingual alternatives [19]. Evaluation frameworks must assess fairness alongside accuracy [12]. The strong prompting effect suggests bias mitigation through prompt engineering may be more cost-effective than model retraining for low-resource languages.

Our findings complement recent Greek-specific work: while Gkovedarou et al. [6] documented systematic masculine defaults in machine translation, and Hackenbuchner et al. [7] established cross-linguistic benchmarks, we demonstrate that coreference resolution in LLMs exhibits distinct bias patterns amenable to prompting-based mitigation. Future work should integrate these approaches, adapting counterfactual data augmentation methods [22] and parallel corpus methodologies [1] for comprehensive Greek gender bias evaluation.

Over-specification Hurts. Complex prompting strategies (Detailed, 0.3%; Chain-of-Thought, 8.4%) dramatically underperform simpler approaches (Few-shot, 68.3%; Neutral, 50.5%). Detailed prompts providing explicit linguistic analysis achieve near-zero ambiguity recognition despite—or perhaps because of—their comprehensiveness. This aligns with recent findings that over-specification can confuse models on straightforward tasks, suggesting that clarity and example-based learning outweigh grammatical explicitness for ambiguity recognition.

Instruction-Following Confound. An important caveat concerns Both% as a measure of genuine ambiguity recognition. Models producing higher Both% scores may do so partly because they are better at following instructions generally, rather than because they have deeper understanding of grammatical ambiguity. Disentangling these factors remains an important direction for future work.

7 Conclusion

We presented the first gender bias evaluation for Greek pronoun resolution in LLMs. Our epicene noun dataset provides a replicable methodology for morphologically-rich languages [21,2]. Our results suggest patterns consistent with a performance-fairness trade-off, with language-specific models showing higher ambiguity recognition alongside larger gender inconsistency. Few-shot prompting appears to substantially mitigate this trade-off (68.3% recognition, 5.0% bias) without model retraining, suggesting that strategic prompt engineering may be more effective than model selection for bias mitigation in low-resource languages — though further work with more models is needed to confirm this generalization.

Limitations

Binary gender. We examine only masculine/feminine pronouns, excluding neuter forms and non-binary identities [4].
Constructed sentences. The dataset uses synthetic rather than naturally-occurring text [10].
Model coverage. We evaluate four LLMs; rapid evolution may require updates.
Greek prompts. Greek-language prompting was tested only on Krikri, limiting cross-model comparison.
Correlation interpretation. The performance-fairness correlation (r=0.89, p=0.109) is based on only four models and is not statistically significant, limiting generalizability.

Acknowledgments. This work was supported by the project “Human-centred AI for a Sustainable and Adaptive Society” (reg. no.: CZ.02.01.01/00/23_025/0008691), co-funded by the European Union, and by GAUK Project No. 324426.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Alhafni, B., Habash, N., Bouamor, H.: The arabic parallel gender corpus 2.0: Extensions and analyses. In: Proceedings of the Thirteenth Language Resources and Evaluation Conference. pp. 1870–1884. European Language Resources Association, Marseille, France (Jun 2022)
2. Bartl, M., Nissim, M., Gatt, A.: Unmasking contextual stereotypes: Measuring and mitigating BERT's gender bias. In: Proceedings of the Second Workshop on Gender Bias in Natural Language Processing. pp. 1–16. Association for Computational Linguistics, Barcelona, Spain (Online) (Dec 2020)
3. Blodgett, S.L., Barocas, S., Daumé III, H., Wallach, H.: Language (technology) is power: A critical survey of "bias" in NLP. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. pp. 5454–5476. Association for Computational Linguistics, Online (Jul 2020). https://doi.org/10.18653/v1/2020.acl-main.485
4. Cao, Y.T., Daumé III, H.: Toward gender-inclusive coreference resolution. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. pp. 4568–4595. Association for Computational Linguistics, Online (Jul 2020). https://doi.org/10.18653/v1/2020.acl-main.418
5. Gallegos, I.O., Rossi, R.A., Barrow, J., Tanjim, M.M., Kim, S., Dernoncourt, F., Yu, T., Zhang, R., Ahmed, N.K.: Bias and fairness in large language models: A survey. Computational Linguistics **50**(3), 1097–1179 (2024). https://doi.org/10.1162/coli_a_00524
6. Gkovedarou, K., Daems, J., De Bruyne, L.: Gender bias in English-to-Greek machine translation. In: Proceedings of the Workshop on Gender-Inclusive Translation Technologies (GITT). European Association for Machine Translation, Geneva, Switzerland (Jun 2025), to appear
7. Hackenbuchner, J., Gkovedarou, K., Daems, J.: GENDEROUS: Machine translation and cross-linguistic evaluation of a gender-ambiguous dataset. In: Proceedings of the 5th Workshop on Gender Bias in Natural Language Processing (GeBNLP). Association for Computational Linguistics, Bangkok, Thailand (Aug 2025), to appear at ACL 2025
8. Ilsp Research Team: Llama-krikri: A greek language model. https://huggingface.co/ilsp/Llama-Krikri-8B (2024), greek-optimized LLM trained on 56.7B tokens
9. Kaneko, M., Imankulova, A., Bollegala, D., Okazaki, N.: Gender bias in masked language models for multiple languages. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 2740–2750. Association for Computational Linguistics, Seattle, United States (Jul 2022). https://doi.org/10.18653/v1/2022.naacl-main.197
10. Levy, S., Lazar, K., Stanovsky, G.: Collecting a large-scale gender bias dataset for coreference resolution and machine translation. In: Findings of the Association for Computational Linguistics: EMNLP 2021. pp. 2470–2480. Association for Computational Linguistics, Punta Cana, Dominican Republic (Nov 2021). https://doi.org/10.18653/v1/2021.findings-emnlp.211
11. Miltsakaki, E.: Toward an anaphor resolution strategy for expository texts in Modern Greek. In: Proceedings of the 4th Discourse Anaphora and Anaphor Resolution Colloquium (DAARC). pp. 113–118 (2002)
12. Orgad, H., Belinkov, Y.: Choose your lenses: Flaws in gender bias evaluation. In: Proceedings of the 4th Workshop on Gender Bias in Natural Language Process-

ing (GeBNLP). pp. 151–167. Association for Computational Linguistics, Seattle, Washington (Jul 2022). https://doi.org/10.18653/v1/2022.gebnlp-1.17
13. Roussis, D., Voukoutis, L., Paraskevopoulos, G., Sofianopoulos, S., Prokopidis, P., Papavasileiou, V., Katsamanis, A., Piperidis, S., Katsouros, V.: Krikri: Advancing open large language models for greek. arXiv preprint arXiv:2505.13772 (2025)
14. Rudinger, R., Naradowsky, J., Leonard, B., Van Durme, B.: Gender bias in coreference resolution. In: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers). pp. 8–14. Association for Computational Linguistics, New Orleans, Louisiana (Jun 2018). https://doi.org/10.18653/v1/N18-2002
15. Stańczak, K., Augenstein, I.: A survey on gender bias in natural language processing. arXiv preprint arXiv:2112.14168 (2021), https://arxiv.org/abs/2112.14168
16. Sun, T., Gaut, A., Tang, S., Huang, Y., ElSherief, M., Zhao, J., Mirza, D., Belding, E., Chang, K.W., Wang, W.Y.: Mitigating gender bias in natural language processing: Literature review. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. pp. 1630–1640. Association for Computational Linguistics, Florence, Italy (Jul 2019). https://doi.org/10.18653/v1/P19-1159
17. Vashishtha, S., Ahuja, K., Sitaram, S.: On evaluating and mitigating gender biases in multilingual settings. In: Findings of the Association for Computational Linguistics: ACL 2023. pp. 5365–5379. Association for Computational Linguistics, Toronto, Canada (Jul 2023). https://doi.org/10.18653/v1/2023.findings-acl.333
18. Webster, K., Recasens, M., Axelrod, V., Baldridge, J.: Mind the GAP: A balanced corpus of gendered ambiguous pronouns. Transactions of the Association for Computational Linguistics **6**, 605–617 (2018). https://doi.org/10.1162/tacl_a_00240
19. Zhao, J., Mukherjee, S., Hosseini, S., Chang, K.W., Awadallah, A.H.: Gender bias in multilingual embeddings and cross-lingual transfer. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. pp. 2896–2907. Association for Computational Linguistics, Online (Jul 2020). https://doi.org/10.18653/v1/2020.acl-main.260
20. Zhao, J., Wang, T., Yatskar, M., Ordonez, V., Chang, K.W.: Gender bias in coreference resolution: Evaluation and debiasing methods. In: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers). pp. 15–20. Association for Computational Linguistics, New Orleans, Louisiana (Jun 2018). https://doi.org/10.18653/v1/N18-2003
21. Zhou, P., Shi, W., Zhao, J., Huang, K.H., Chen, M., Cotterell, R., Chang, K.W.: Examining gender bias in languages with grammatical gender. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). pp. 5276–5284. Association for Computational Linguistics, Hong Kong, China (Nov 2019). https://doi.org/10.18653/v1/D19-1531
22. Zmigrod, R., Mielke, S.J., Wallach, H., Cotterell, R.: Counterfactual data augmentation for mitigating gender stereotypes in languages with rich morphology. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. pp. 1651–1661. Association for Computational Linguistics, Florence, Italy (Jul 2019). https://doi.org/10.18653/v1/P19-1161